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The Log of Gravity

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1. Introduction

Economists have long been aware that Jensen's inequality implies that $E(\ln y) \neq \ln E(y)$, that is, the expected value of the logarithm of a random variable is different from the logarithm of its expected value. This basic fact, however, has been neglected in many econometric applications. Indeed, one important implication of Jensen's inequality is that the standard practice of interpreting the parameters of log-linearized models estimated by ordinary least squares (OLS) as elasticities can be highly misleading in the presence of heteroskedasticity.

Although many authors have addressed the problem of obtaining consistent estimates of the conditional mean of the dependent variable when the model is estimated in the log-linear form (see, for example, Goldberger, 1968; Manning and Mullahy, 2001), we were unable to find any reference in the literature to the potential bias of the elasticities estimated using the log-linear model.

In this paper we use the gravity equation for trade as a particular illustration of how the bias arises and propose an appropriate estimator. We argue that the gravity equation, and, more generally, constant-elasticity models, should be estimated in their multiplicative form and propose a simple pseudo-maximum likelihood (PML) estimation technique. Besides being consistent in the presence of heteroskedasticity, this method also provides a natural way to deal with zero values of the dependent variable.

Using Monte Carlo simulations, we compare the performance of our estimator with that of OLS (in the log-linear specification). The results are striking. In the presence of heteroskedasticity, estimates obtained using log-linearized models are severely biased, distorting the interpretation of the model. These biases might be critical for the comparative assessment of competing economic theories, as well as for the evaluation of the effects of different policies. In contrast, our method is robust to the different patterns of heteroskedasticity considered in the simulations.

We next use the proposed method to provide new estimates of the gravity equation in cross sectional data. Using standard tests, we show that heteroskedasticity is indeed a

severe problem, both in the traditional gravity equation introduced by Tinbergen (1962), and in a gravity equation that takes into account multilateral resistance terms or fixed effects, as suggested by Anderson and van Wincoop (2003). We then compare the estimates obtained with the proposed PML estimator with those generated by OLS in the log-linear specification, using both the traditional and the fixed-effects gravity equations.

Our estimation method paints a very different picture of the determinants of international trade. In the traditional gravity equation, the coefficients on GDP are not, as generally estimated, close to one. Instead, they are significantly smaller, which might help reconcile the gravity equation with the observation that the trade-to-GDP ratio decreases with total GDP (or, in other words, that smaller countries tend to be more open to international trade). In addition, OLS greatly exaggerates the roles of colonial ties, geographical proximity, and bilateral trade agreements.

Using the Anderson-van Wincoop (2003) gravity equation, we find that OLS yields significantly larger effects for geographical distance. The estimated elasticity obtained from the log-linearized equation is almost twice as large as that predicted by PML. OLS also predicts a large role for common colonial ties, implying that sharing a common colonial history practically doubles bilateral trade. In contrast, the proposed PML estimator leads to a statistically and economically insignificant effect.

The general message is that, even controlling for fixed effects, the presence of heteroskedasticity can generate strikingly different estimates when the gravity equation is log-linearized, rather than estimated in levels. In other words, Jensen's inequality is quantitatively and qualitatively important in the estimation of gravity equations. This suggests that inferences drawn on log-linearized regressions can produce misleading conclusions.

Despite the focus on the gravity equation, our criticism to the conventional practice and the solution we propose extends to a broad range of economic applications where the equations under study are log-linearized, or, more generally, transformed by a non-linear function. A short list of examples includes the estimation of Mincerian equations

representation of the constant-elasticity model is only useful as a device to estimate the parameters of interest under very specific conditions on the error term.

When η_i is statistically independent of x_i , the conditional variance of y_i (and ε_i) is proportional to $\exp(2x_i\beta)$. Although economic theory generally does not provide any information on the variance of ε_i , we can infer some of its properties from the characteristics of the data. Because y_i is non-negative, when $E[y_i|x]$ approaches zero, the probability of y_i being positive must also approach zero. This implies that $V[y_i|x]$, the conditional variance of y_i , tends to vanish as $E[y_i|x]$ passes to zero.⁸ On the other hand, when the expected value y is far away from its lower bound, it is possible to observe large deviations from the conditional mean in either direction, leading to greater dispersion. Thus, in practice, ε_i will generally be heteroskedastic and its variance will depend on $\exp(x_i\beta)$, but there is no reason to assume that $V[y_i|x]$ is proportional to $\exp(2x_i\beta)$. Therefore, in general, regressing $\ln(y_i)$ on x_i by OLS will lead to inconsistent estimates of β .

It may be surprising that the pattern of heteroskedasticity and, indeed, the form of all higher-order moments of the conditional distribution of the error term can affect the consistency of an estimator, rather than just its efficiency. The reason is that the non-linear transformation of the dependent variable in equation (7) changes the properties of the error term in a non-trivial way since the conditional expectation of $\ln(\eta_i)$ depends on the shape of the conditional distribution of η_i . Hence, unless very strong restrictions are imposed on the form of this distribution, it is not possible to recover information about the conditional expectation of y_i from the conditional mean of $\ln(y_i)$ simply because $\ln(\eta_i)$ is correlated with the regressors. Nevertheless, estimating (7) by OLS will produce consistent estimates of the parameters of $E[\ln(y_i)|x]$ as long as $E[\ln(y_i)|x]$ is a linear function of

⁸In the case of trade data, when $E[y_i|x]$ is close to its lower bound (i.e., for pairs of small and distant countries), it is unlikely that large values of trade are observed since they cannot be offset by equally large deviations in the opposite direction simply because trade cannot be negative. Therefore, for these observations, dispersion around the mean tends to be small.